

MATH70013 Advanced Simulation Methods

Coursework 2

CID: 06060027

1 Filtering and smoothing for a nonlinear state space model

Model

The state-space model is

$$X_n = \frac{1}{2}X_{n-1} + \frac{25X_{n-1}}{1 + X_{n-1}^2} + 8 \cos(1.2n) + V_n, \quad V_n \sim \mathcal{N}(0, 0.1), \quad (1)$$

$$Y_n = \frac{X_n^2}{20} + W_n, \quad W_n \sim \mathcal{N}(0, 1), \quad (2)$$

with $X_0 \sim \mathcal{N}(0, 1)$ and $T = 200$. One synthetic dataset $(x_{0:T}^*, y_{0:T})$ is simulated and stored for reproducibility.

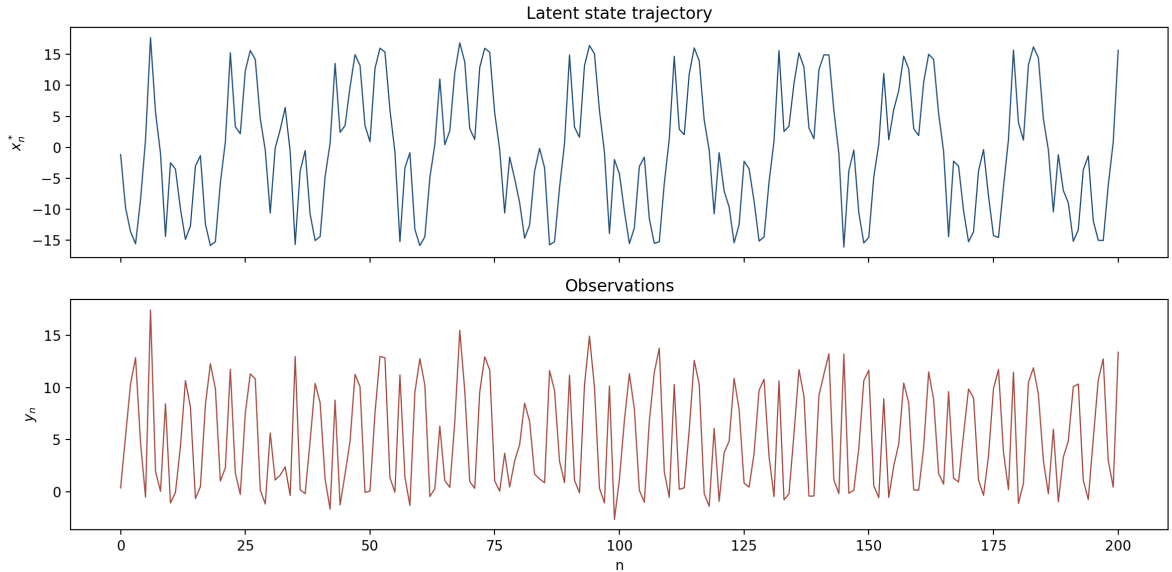


Figure 1: Simulated latent state trajectory x_n^* and corresponding observations y_n . The nonlinear observation has conditional mean $X_n^2/20$, so large $|x_n|$ tends to produce larger observations, although the Gaussian noise can still make individual y_n negative.

1.1 Importance density and smoother choice

(a) SIR particle filter with bootstrap proposal. For the importance density I use the **bootstrap (transition prior) proposal**:

$$q(x_n | y_n, x_{n-1}) = f(x_n | x_{n-1}) = \mathcal{N}(x_n; \mu_f(x_{n-1}, n), 0.1),$$

where $\mu_f(x, n) = \frac{1}{2}x + \frac{25x}{1+x^2} + 8 \cos(1.2n)$.

Choice of proposal. The bootstrap proposal is simple and stable for this highly nonlinear model, where no closed-form locally optimal proposal exists. However, because $Y_n = X_n^2/20 + W_n$

depends on X_n^2 , the observation is symmetric in x_n (both $+x_n$ and $-x_n$ produce similar y_n), which can make the posterior bimodal. The bootstrap proposal ignores y_n entirely when proposing, so when y_n is very informative (large y_n , constraining $|x_n|$), many proposed particles may land in low-likelihood regions, producing weight degeneracy. I monitor this through ESS.

Systematic resampling is applied at every step, which has $O(N)$ cost and lower variance than multinomial resampling. In this first part I only use the filter for marginal filtering and smoothing approximations, so I do not rely on the particle normalising constant as an unbiased likelihood estimator. In the likelihood and PMMH parts, where likelihood unbiasedness is part of the argument, I use multinomial resampling.

(b) FFBSa (Forward Filtering Backward Sampling). I use **FFBSa** with $M = N$ backward trajectories. At each backward step $n = T - 1, \dots, 0$, for each of the M backward samples $x_{n+1}^{(m)}$, the backward weight for particle i at time n is

$$\tilde{w}_n^{(i)} \propto w_n^{(i)} f(x_{n+1}^{(m)} | x_n^{(i)}),$$

and we draw one index from these weights. This produces M approximately independent smoothed trajectories from $p(x_{0:T} | y_{0:T})$, in contrast to the forward-path approximation which suffers genealogy collapse.

The computational cost is $O(NMT)$ (here $O(N^2T)$ since $M = N$), which is more expensive than the forward PF ($O(NT)$) but feasible for $N \leq 500$ and $T = 200$.

Density plots. Figures 2 and 3 compare the filter approximation to $p(x_n | y_{0:n})$, the genealogy-based naive smoother, and the FFBSa smoother¹ at $n \in \{1, 50, 100, 150, 199\}$.

¹Here “smoother” refers to the conditioning set rather than visual smoothness: the filter targets $p(x_n | y_{0:n})$, whereas the smoother targets $p(x_n | y_{0:T})$. In this nonlinear model the smoothing marginal can be more peaked, shifted, or multimodal when future observations resolve the sign ambiguity in $Y_n = X_n^2/20 + W_n$.

Density approximations, $N=50$

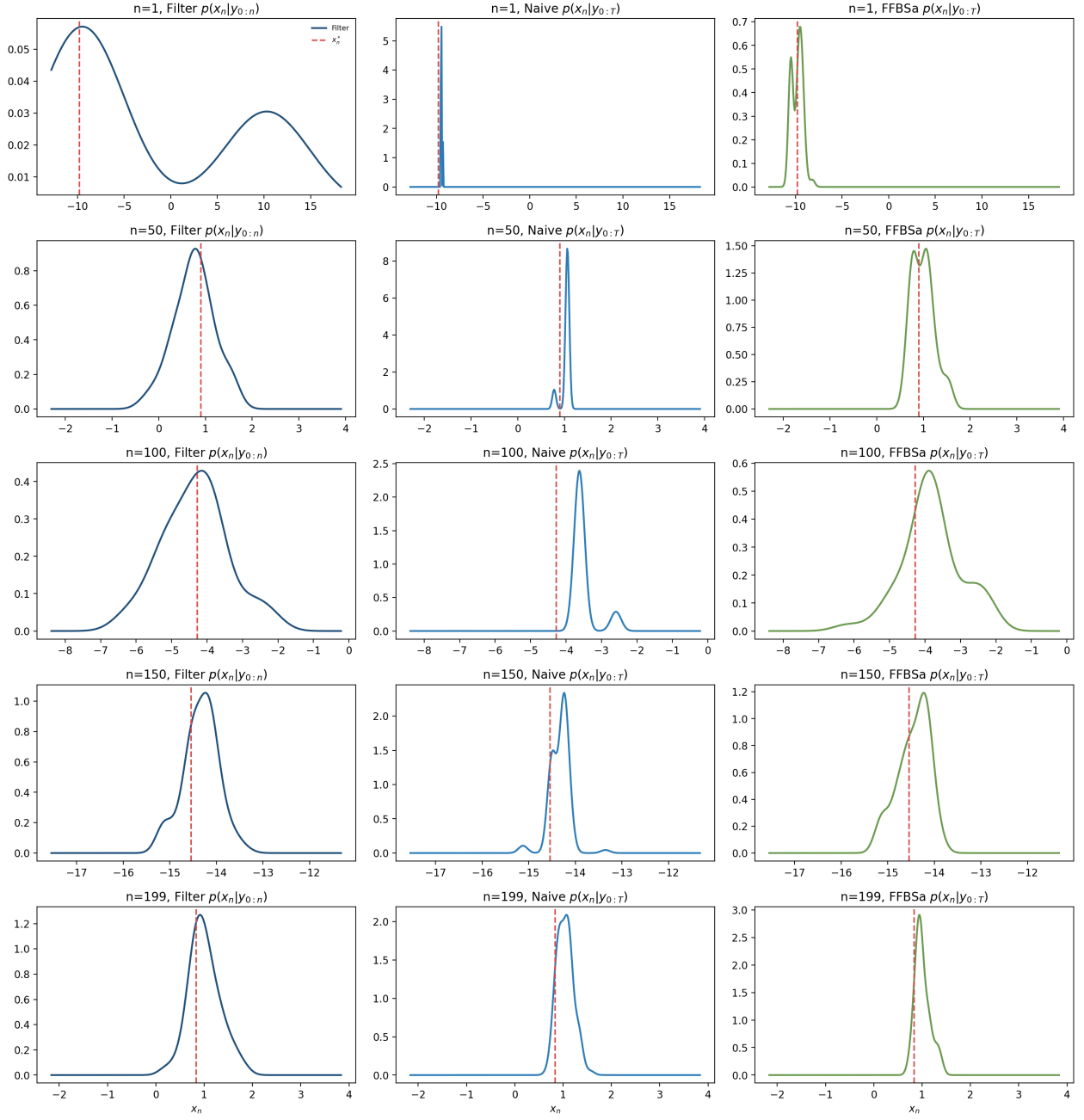


Figure 2: Density approximations at snapshot times, $N = 50$. With only 50 particles, the filter densities are rough and the naive smoother shows clear signs of path degeneracy; at early times, in particular, the naive smoother distribution can collapse to a few atoms. FFBSa avoids this atomic collapse and gives a less degenerate Monte Carlo representation of the smoothing marginal.

Density approximations, $N=500$

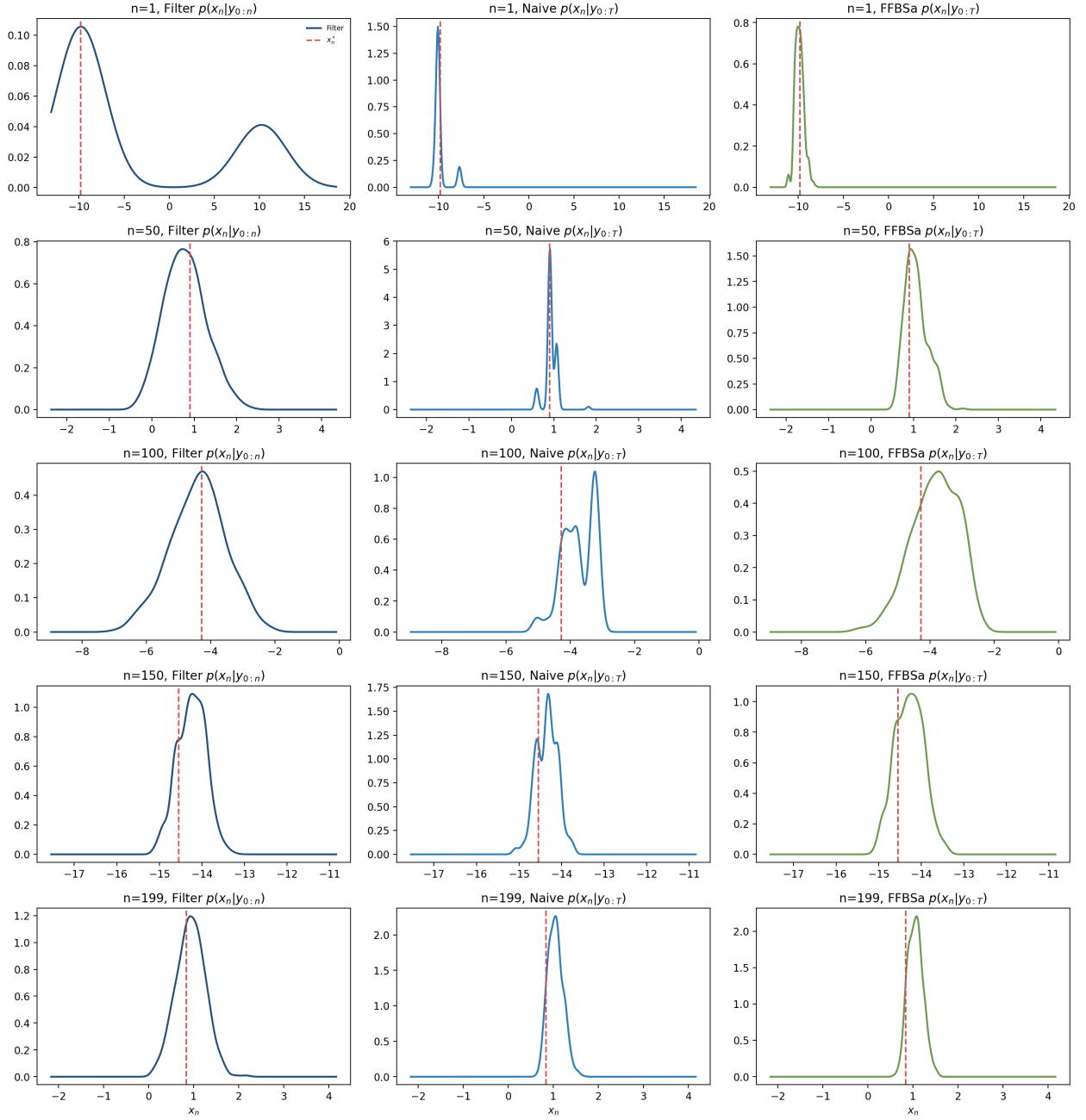


Figure 3: Density approximations at snapshot times, $N = 500$. With more particles, all three KDE estimates are less noisy. The filter and FFBSa smoothing marginal are visually similar at many times, but smoothing can shift mass when future observations are informative.

Density interpretation. With $N = 50$ the filter densities are noisy but capture the general location of x_n^* . The filter and FFBSa panels are often close in these snapshots, especially away from the first time point, so the smoothing correction is not visually dramatic at every displayed n . This is expected: for many of these times the filtering marginal is already quite concentrated. The clearest difference is instead the behaviour of the naive forward-path smoother. At early times, especially $n = 1$, it collapses to very few unique values due to path degeneracy, because the final-time genealogy has coalesced and the “smoothing” distribution at past times is degenerate. FFBSa avoids this by independently sampling backward paths.

Comparing $N = 50$ and $N = 500$: the latter produces less noisy KDE approximations and captures multimodality more reliably. The observation $Y_n = X_n^2/20 + W_n$ creates sign ambiguity,

since both $+x$ and $-x$ can explain the same y_n . In the displayed snapshots this is most visible at $n = 1$, where the filtering density is clearly bimodal.

1.2 Path degeneracy

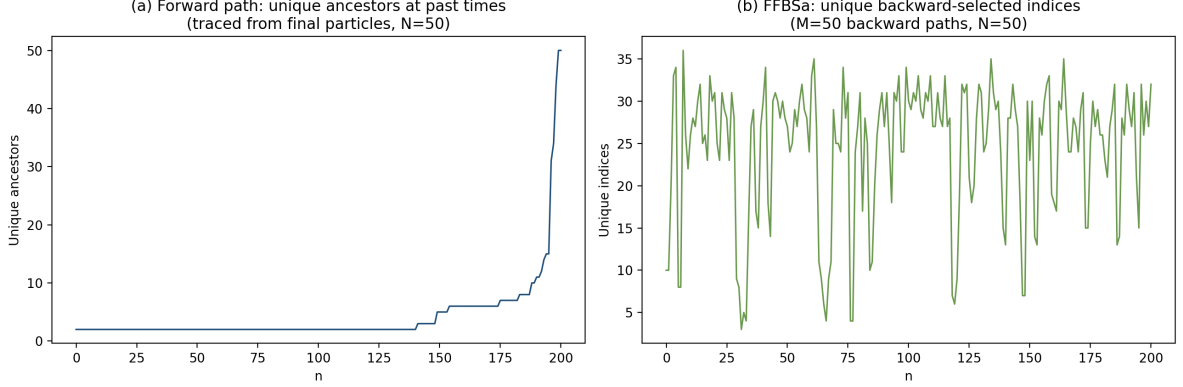


Figure 4: Path degeneracy diagnostics for $N = 50$. **Left:** number of unique ancestors at each past time when tracing the genealogy of final-time particles backward. This collapses rapidly as we move away from T , reaching only a few unique ancestors for early and middle times. **Right:** number of unique backward-selected particle indices at each time among $M = 50$ FFBSa trajectories.

Path degeneracy. The left panel of Figure 4 demonstrates path degeneracy: the ancestral genealogy coalesces backward in time, so at early times ($n \ll T$) the forward-path approximation to $p(x_{0:T} | y_{0:T})$ effectively reduces to a few distinct trajectories. Path degeneracy is distinct from weight degeneracy (measured by ESS): even when the marginal filter $p(x_n | y_{0:n})$ is well approximated at each n , the *joint* smoothing distribution carried by the genealogy is degenerate.

The right panel shows that FFBSa maintains substantially higher path diversity, measured by counting unique backward-selected particle indices at each time among $M = 50$ trajectories. More carefully, FFBSa still relies on the same filtering cloud, but it avoids the *severe* early-time collapse of the naive genealogy-based smoother because it samples backward from approximate conditional distributions rather than tracing only terminal genealogies. This is why dedicated smoothing algorithms are needed for quantities that depend on trajectories rather than marginals, such as the additive functional studied below.

1.3 Mean and variance trajectories

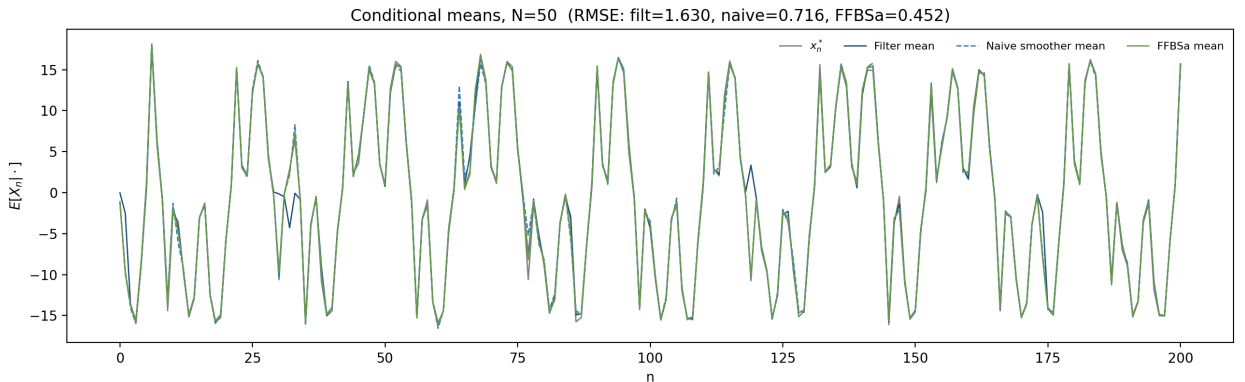


Figure 5: Conditional means for $N = 50$. RMSE values are shown in the title.

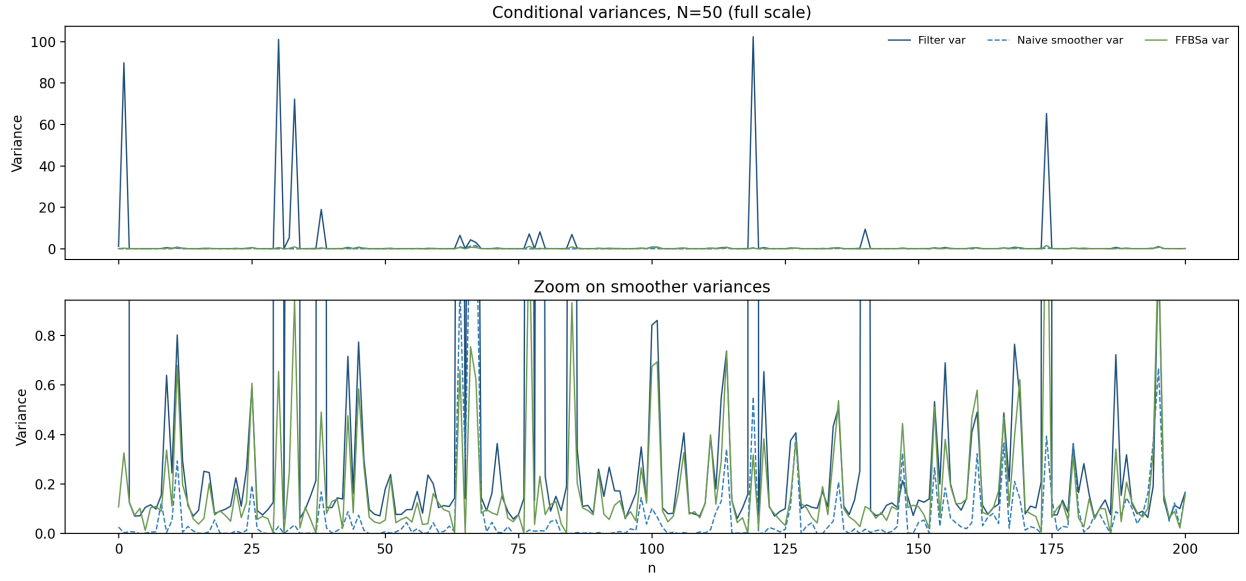


Figure 6: Conditional variances for $N = 50$. The lower panel zooms in on the naive smoother and FFBSa variance curves.

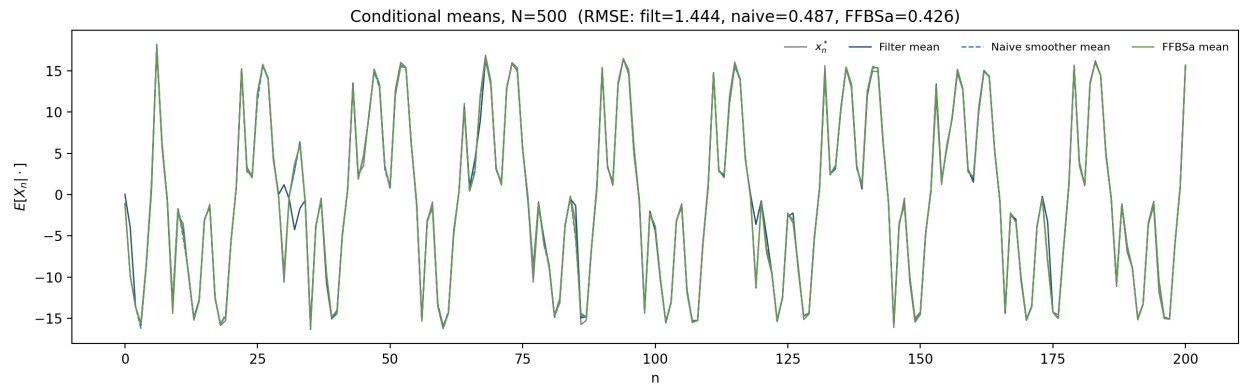


Figure 7: Conditional means for $N = 500$. RMSE values are shown in the title.

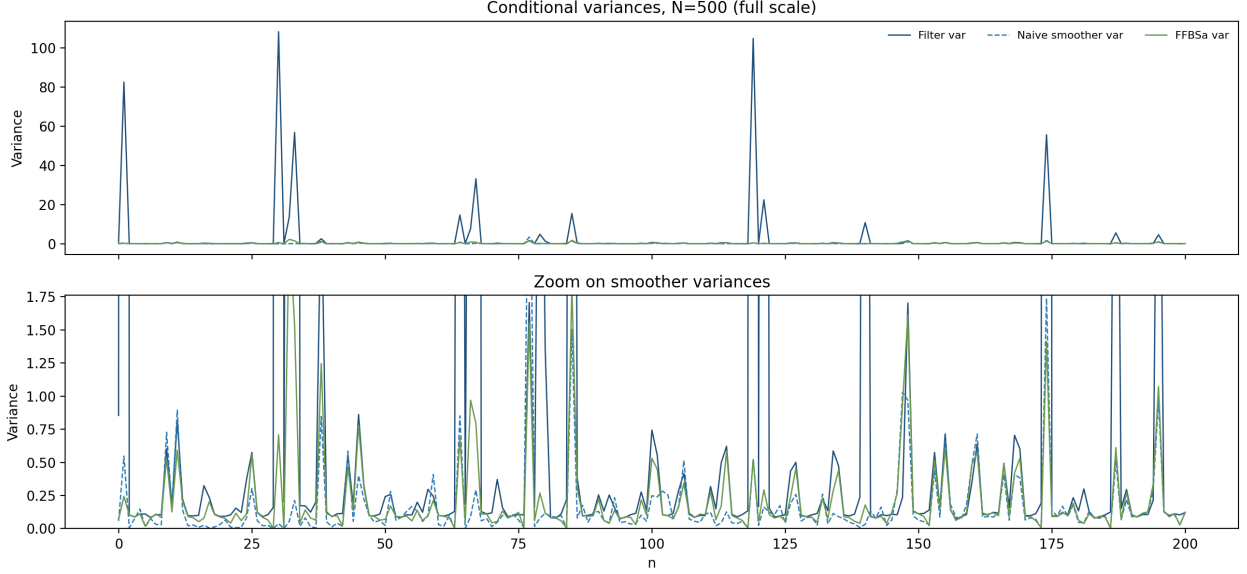


Figure 8: Conditional variances for $N = 500$. The lower panel zooms in on the naive smoother and FFBSa variance curves.

Table 1: Summary statistics for Q1.

	$N = 50$			$N = 500$		
	Filter	Naive	FFBSa	Filter	Naive	FFBSa
RMSE	1.630	0.716	0.452	1.444	0.487	0.426
Avg. Var	2.683	0.073	0.183	2.907	0.175	0.221

1.4 Discussion

Estimation performance. Table 1 shows that FFBSa achieves the lowest RMSE in both particle regimes: 0.452 at $N = 50$ and 0.426 at $N = 500$. The naive forward-path smoother improves over the filter (RMSE 0.716 vs. 1.630 at $N = 50$), but it remains less accurate than FFBSa because genealogy collapse degrades early-time smoothing estimates. Increasing N to 500 helps all three methods: filter RMSE drops from 1.630 to 1.444, and naive smoother RMSE drops from 0.716 to 0.487. However, this does not eliminate the structural path-degeneracy issue in the naive smoother; FFBSa still provides an additional reduction (from 0.487 to 0.426) even at $N = 500$.

The naive smoother’s average variance (0.073 at $N = 50$) is anomalously low, not because it is accurate, but because path degeneracy reduces the particle cloud to very few distinct trajectories, artificially shrinking the empirical variance. FFBSa’s variance (0.183) is larger but more representative of the smoothing posterior spread than the path-degenerate naive smoother estimate.

The average filter variance is slightly larger at $N = 500$ than at $N = 50$ (2.907 vs. 2.683). This is not a contradiction: the variance is an estimate of posterior spread, not an estimation error. With only 50 particles, particle impoverishment and missed modes can underestimate the true filtering spread, while 500 particles better resolve the broad or bimodal filtering distributions.

Filtering vs. smoothing. Filtering provides $p(x_n | y_{0:n})$, the online posterior using only past and current data. Smoothing gives $p(x_n | y_{0:T})$, incorporating future observations. In this model, future data are particularly helpful because the squared observation map loses the sign of x_n ; subsequent time steps, through the autoregressive dynamics, can disambiguate.

Role of N and model features. The process noise variance $\sigma_V^2 = 0.1$ is small relative to the observation noise $\sigma_W^2 = 1$, so the transition is relatively tight and the bootstrap proposal (which samples from the transition) is not unreasonable. The average ESS is 39.9 at $N = 50$ and 399.9 at $N = 500$, indicating that systematic resampling at every step keeps the weights well-behaved. The nonlinear drift $25x/(1+x^2)$ and periodic forcing $8\cos(1.2n)$ give the latent state structured variation; the squared observation creates sign ambiguity, visible most clearly at $n = 1$ in Figures 2 and 3.

Computational cost. The bootstrap PF costs $O(NT)$; FFBSa with $M = N$ costs $O(N^2T)$. For $N = 500$, $T = 200$, this is 5×10^7 transition density evaluations in the backward pass, which is feasible but noticeably slower than filtering alone.

1.5 Theoretical property for additive functional

The target is the smoothed additive functional

$$S_n^\theta = \int \left[\sum_{k=0}^n x_k \right] p_\theta(x_{0:n} \mid y_{0:n}) dx_{0:n} = \mathbb{E}_\theta \left[\sum_{k=0}^n X_k \mid y_{0:n} \right].$$

(a) Standard SIR path estimator. The SIR particle filter carries trajectories through the genealogy. The estimator is

$$\hat{S}_n^{\text{SIR}} = \sum_{i=1}^N w_n^{(i)} \sum_{k=0}^n x_k^{b_k^{(i)}},$$

where $b_k^{(i)}$ traces the ancestor of particle i at time n back to time k . For path-space particle estimates of additive functionals, the variance growth is quadratic in n . In standard formulations this is expressed through matching upper and lower rates, for example

$$\text{Var}[\hat{S}_n^{\text{SIR}}] = O(n^2/N) \quad \text{and, in path-degenerate regimes,} \quad \text{Var}[\hat{S}_n^{\text{SIR}}] = \Omega(n^2/N),$$

so the rate is sharp up to constants. The mechanism is path degeneracy: the sum $\sum_{k=0}^n x_k$ requires the full trajectory, but the genealogy collapses, so effectively only a few distinct trajectory prefixes contribute.

(b) Smoother-based estimator. Since the dedicated smoother selected above is FFBSa, the corresponding forward-backward smoothing result gives linear growth in n : under standard forgetting/mixing assumptions, particle smoothing estimators of S_n^θ based on FFBSa/FFBSm satisfy bias and variance bounds of order n/N ,

$$\mathbb{E}[\hat{S}_n^{\text{sm}} - S_n^\theta] = O(n/N), \quad \text{Var}[\hat{S}_n^{\text{sm}}] = O(n/N).$$

For the numerical verification below, I use FFBSa directly. For each terminal time n , sample M backward trajectories from the particle approximation of $p(x_{0:n} \mid y_{0:n})$. Start with $J_n^{(m)}$ drawn from the filtering weights $w_n^{(i)}$, then sample backward indices using the pair-kernel

$$\mathbb{P}(J_k^{(m)} = j \mid J_{k+1}^{(m)} = i, y_{0:k}) = \frac{w_k^{(j)} f(x_{k+1}^{(i)} \mid x_k^{(j)})}{\sum_{\ell=1}^N w_k^{(\ell)} f(x_{k+1}^{(i)} \mid x_k^{(\ell)})}.$$

The direct FFBSa estimator is then

$$\hat{S}_n^{\text{FFBSa}} = \frac{1}{M} \sum_{m=1}^M \sum_{k=0}^n x_k^{J_k^{(m)}}.$$

Crucially, this is applied separately to each prefix $y_{0:n}$, so the estimator targets exactly the quantity in the question rather than the full-horizon smoother $p(x_{0:n} | y_{0:T})$. Taking $M = N$ gives the same practical linear target:

$$\text{Var}[\hat{S}_n^{\text{FFBSa}}] = O(n/N),$$

i.e. linear in n rather than quadratic. Computing this for all $n = 0, \dots, T$ is more expensive than filtering because FFBSa must be rerun for each prefix, but it is feasible here with $T = 200$.

Verification procedure. Fix the dataset $(x_{0:T}^*, y_{0:T})$. Following the assignment budget, I use $R = 50$ independent runs of each algorithm (each with a different random seed):

1. Run the SIR PF with N particles, compute $\hat{S}_n^{\text{SIR}}$ for $n = 0, \dots, T$.
2. Using the same filtering particles, apply direct FFBSa with $M = N$ to each prefix $y_{0:n}$ and compute $\hat{S}_n^{\text{FFBSa}}$ for $n = 0, \dots, T$.
3. Compute the empirical variance $\widehat{\text{Var}}_R[\hat{S}_n]$ across the R runs for each n .

Plot $\widehat{\text{Var}}$ vs. n , and the scaled versions $\widehat{\text{Var}}/n$ and $\widehat{\text{Var}}/n^2$. Also fit $\log \widehat{\text{Var}} = a + b \log n$ and report the slope b : the path-space SIR estimator should show faster-than-linear growth, while the direct FFBSa estimator should be compatible with an approximately linear-in- n regime.

1.6 Numerical verification

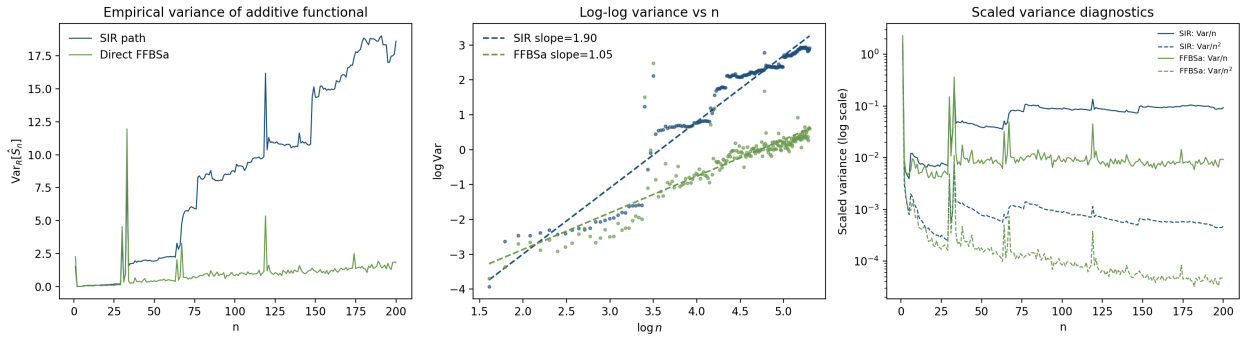


Figure 9: Empirical variance of the additive functional estimator across $R = 50$ runs with $N = M = 150$. **Left:** raw variance. **Centre:** log-log plot with fitted slopes. **Right:** scaled variance Var/n and Var/n^2 on a log scale.

Numerical interpretation. Fitting the log-log slope over $n \geq 5$ gives $b_{\text{SIR}} = 1.90$ and $b_{\text{FFBSa}} = 1.05$. As sensitivity checks, fitting only over $n \geq 20$ gives slopes 1.83 and 0.91, and fitting over $n \geq 50$ gives slopes 1.49 and 0.78. These finite- T estimates should not be read as exact asymptotic exponents, especially under the assignment's limit of only 50 independent runs. The raw variance plot has several finite-sample jumps, so the fitted SIR slope is range-sensitive. However, the pattern is consistent with the theory: the path-space SIR estimator grows much faster than linearly, while the direct FFBSa estimator stays close to a linear growth regime.

The scaled variance plots should be interpreted cautiously: neither curve is perfectly flat at this finite N and T . Still, the direct FFBSa estimator has a much lower variance level. By the final time point, the empirical variance of the SIR path estimator is about 10.12 times that of direct FFBSa.

This has direct practical implications for smoothing-based Monte Carlo approximations: estimates built from terminal genealogies can be much noisier than estimates based on backward-sampled smoothing trajectories.

2 Likelihood inference using synthetic data

Models

$$\text{Model 1: } X_n = \rho X_{n-1} + \tau V_n, \quad Y_n = X_n + \sigma W_n, \quad (3)$$

$$\text{Model 2: } X_n = \rho X_{n-1} + \tau V_n, \quad Y_n = X_n + \sigma \exp(X_n/2) W_n, \quad (4)$$

with $T = 100$, $\rho = 0.8$, $\sigma = 1$, $\tau = 0.5$, $X_0 \sim \mathcal{N}(0, 1)$, and the same latent trajectory $x_{0:T}^*$ for both datasets y^1 (Model 1) and y^2 (Model 2).

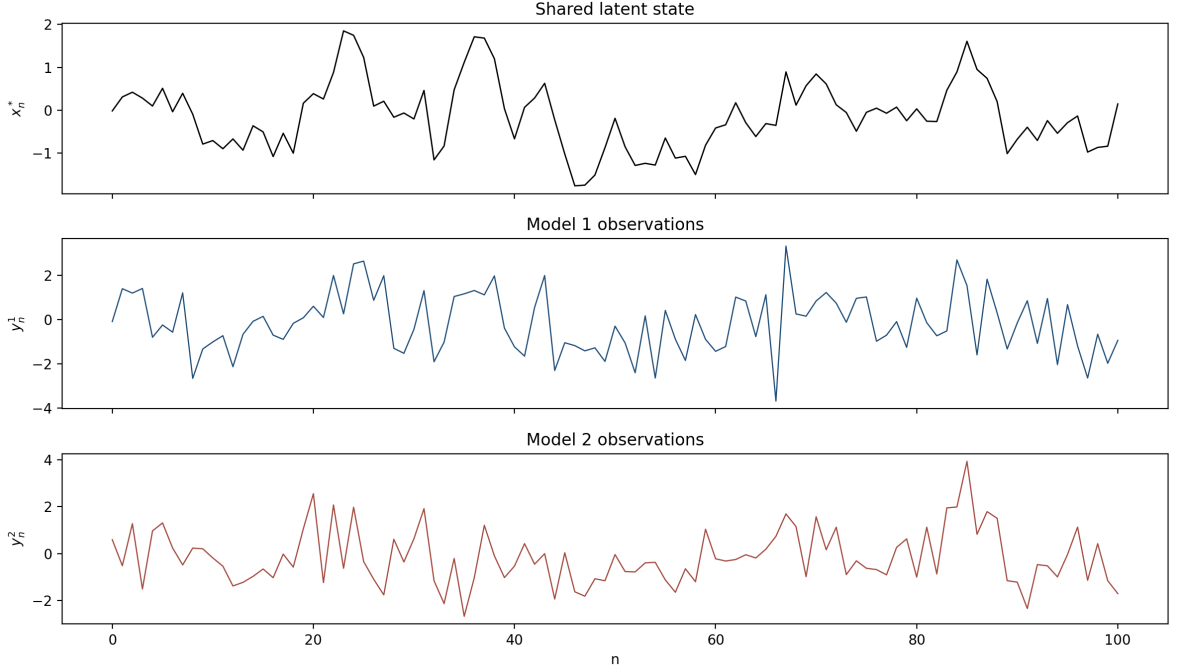


Figure 10: Shared latent state and observations from both models. Model 2 observations have state-dependent variance: positive x_n inflates the noise, while negative x_n shrinks it.

2.1 Likelihood grid

For Model 1, the exact marginal likelihood $p_{\sigma^2,1}(y_{0:T})$ is computed via the Kalman filter. For both models, the particle approximation $\hat{p}_{\sigma^2,j}^N(y_{0:T})$ uses a bootstrap SIR particle filter with $N = 500$ and multinomial resampling over a 200-point grid $\sigma^2 \in [0.01, 2]$. At each grid point I run $R = 10$ independent PF likelihood estimators $\hat{p}^{(r)}$ and plot

$$\log\left(\frac{1}{R} \sum_{r=1}^R \hat{p}^{(r)}\right),$$

i.e. log-mean-exp of the individual log-likelihood estimates. This keeps the averaging on the likelihood scale before taking logs and reduces Monte Carlo roughness in the surface. Multinomial resampling ensures that each per-step normalising constant estimator is unbiased, so the product-form likelihood estimator is itself unbiased.

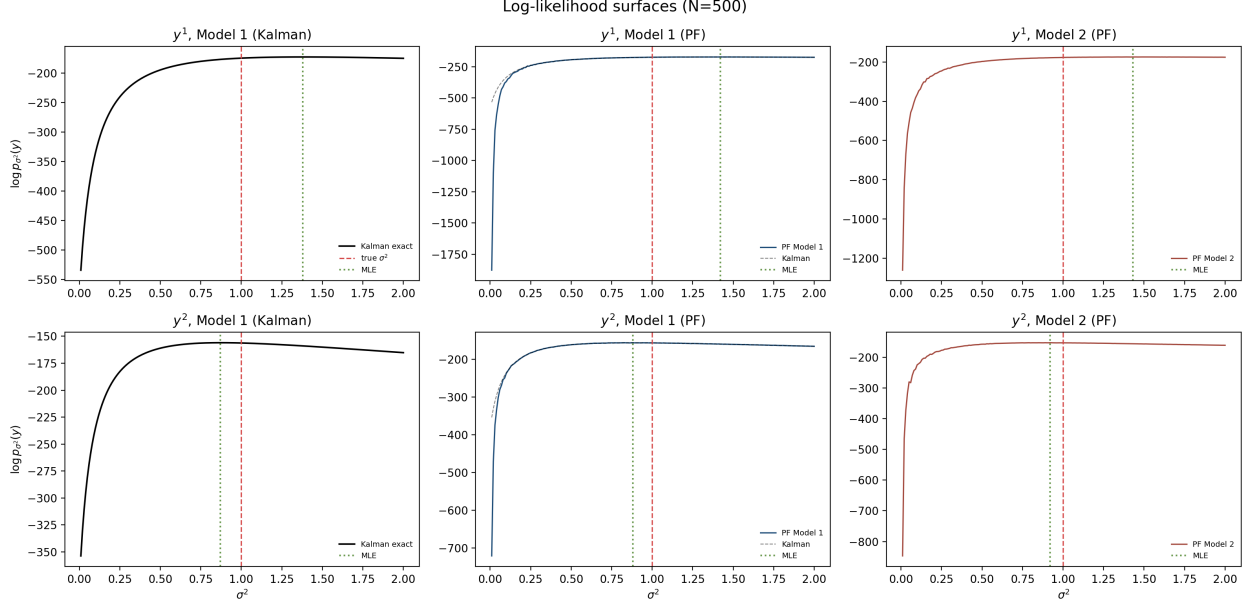


Figure 11: Log-likelihood surfaces. Top row: y^1 ; bottom row: y^2 . Green dotted lines mark the grid MLE; red dashed lines mark $\sigma^2 = 1$.

Table 2: MLE estimates of σ^2 from grid search. True value: $\sigma^2 = 1$.

Setting	Data y^1	Data y^2
Model 1 (Kalman exact)	1.380	0.870
Model 1 (PF, $N = 500$, $R = 10$)	1.420	0.880
Model 2 (PF, $N = 500$, $R = 10$)	1.430	0.920

2.2 Discussion

Intuition from data. Model 1 has constant observation variance σ^2 , so y^1 has roughly uniform noise amplitude around x_n^* . Model 2 has state-dependent variance $\sigma^2 \exp(X_n)$: when x_n is large and positive, the observation noise inflates exponentially; when x_n is negative, the noise shrinks. Since both datasets share the same $x_{0:T}^*$, any difference in y^1 vs. y^2 is entirely due to the observation model.

Observed MLE values and model misspecification. For y^1 (generated by Model 1), the Kalman MLE is $\hat{\sigma}^2 = 1.380$ and the PF MLEs under Models 1 and 2 are 1.420 and 1.430 respectively. The two PF estimates are nearly identical, indicating that with $T = 100$ and a single trajectory the likelihood does not strongly separate the two observation models when the data are actually homoscedastic. The fact that all three estimates are above the true value $\sigma^2 = 1$ should be read as finite-sample variability from one length-100 dataset, not as evidence of systematic bias in the exact Kalman method.

For y^2 (generated by Model 2), the Kalman MLE under Model 1 is 0.870, the PF Model 1 MLE is 0.880, and the PF Model 2 MLE is 0.920, which is closer to the true value $\sigma^2 = 1$. Model 2 is structurally correct here, but the difference between 0.880 and 0.920 is modest: with a single sample of length $T = 100$, model misspecification affects the *interpretation* of $\hat{\sigma}^2$ more clearly than its numerical value. Under Model 1, $\hat{\sigma}^2$ absorbs both the true observation noise and the unmodelled heteroscedasticity into a single parameter.

Particle approximation quality. For Model 1, the PF log-likelihood closely tracks the exact Kalman surface near the high-likelihood region of Figure 11, with small Monte Carlo roughness visible. The agreement is much worse when σ^2 is very close to zero: the PF curve drops far below the Kalman curve because the observations become extremely informative, while the bootstrap proposal still samples from the transition and ignores the current observation. This creates severe weight degeneracy and high variance in the likelihood estimate. The issue is visible in the small- σ^2 left tail of the likelihood plots, although it does not determine the MLE here because the maxima occur away from that boundary.

With multinomial resampling, the SIR particle likelihood estimator $\hat{p}^N$ is nonnegative and unbiased on the likelihood scale, but $\log \hat{p}^N$ is biased downward by Jensen’s inequality. This downward bias is small for $N = 500$ near the likelihood maximum, but it can still shift the MLE slightly: the PF MLE for y^1 under Model 1 is 1.420 vs. the Kalman MLE of 1.380. The $R = 10$ replication averaging through log-mean-exp smooths the surface, but it does not remove the underlying log-bias.

For Model 2, the likelihood curves are flatter around the high-likelihood region, especially for moderate to large σ^2 . Therefore the reported Model 2 grid maximisers should be interpreted as approximate locations on a relatively shallow surface, not as sharply identified optima. This also explains why small Monte Carlo perturbations can move the PF argmax without changing the qualitative model comparison.

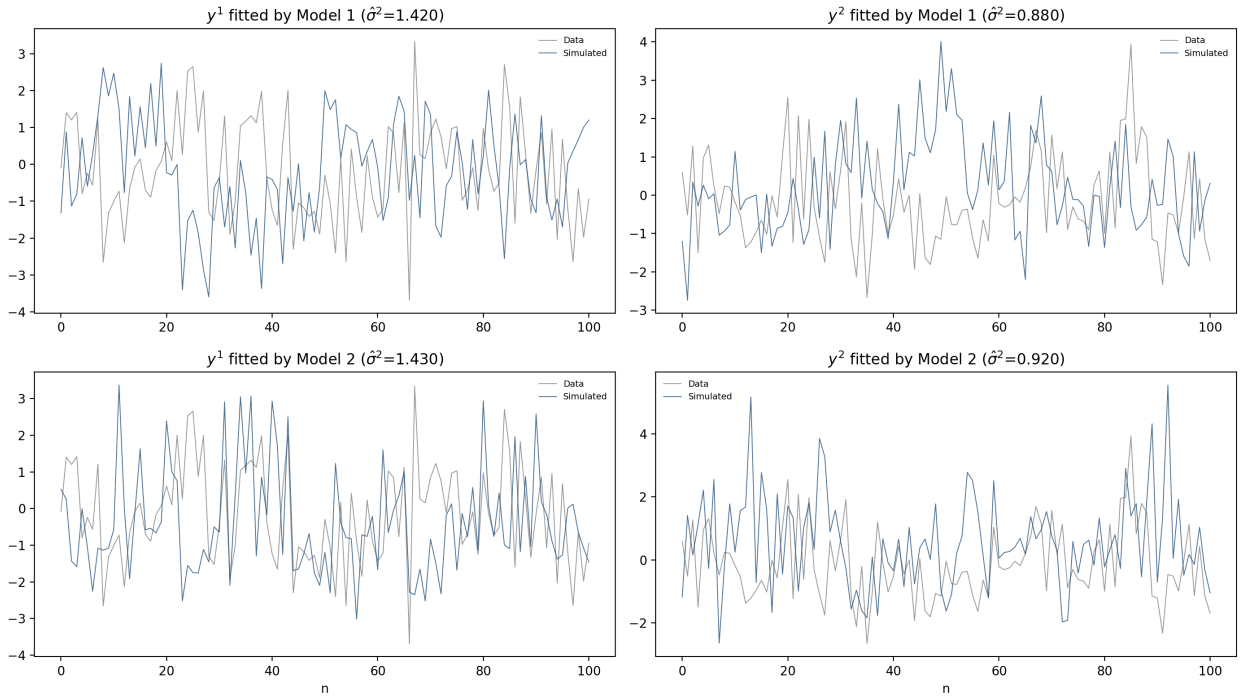


Figure 12: Simulations from fitted models vs. original data. Comparing noise amplitude and heteroscedasticity.

Fitted model simulations. These pathwise simulations are illustrative rather than a formal goodness-of-fit test: a single simulated series is not expected to match the observed path point by point. The purpose is to compare broad scale features such as amplitude and visible heteroscedasticity. Model 1 can only adjust one constant observation variance, so its fitted simulations spread noise roughly uniformly over time. Model 2 changes the conditional observation variance through $\sigma^2 \exp(X_n)$, so positive latent states create wider observation bands and negative latent states create narrower bands. For y^2 , this reproduces the main structural feature of the data more directly: high volatility periods line up with positive parts of the latent trajectory, while quieter periods line up

with negative parts. This explains why the Model 2 MLE for y^2 moves closer to the true $\sigma^2 = 1$ and why its simulations look more plausible. For y^1 , the data are homoscedastic, so this state dependent variance mechanism is not needed; the fitted Model 2 simulations do not give a better qualitative fit, consistent with the similar MLE values in Table 2.

2.3 Refinements

Possible improvements to the grid likelihood method:

1. **Increase N or average over multiple PF runs** per grid point to reduce Monte Carlo noise in the log-likelihood surface. (Already implemented here with $R = 10$.)
2. **Optimise on $\log \sigma^2$ scale:** ensures positivity and often gives a more symmetric likelihood surface, improving grid resolution near zero.
3. **EM or iterative methods** avoid the exhaustive grid search and scale to higher-dimensional parameter spaces where a grid is infeasible.
4. **Finer or adaptive grid:** concentrate grid points near the maximum after a coarse initial sweep.

2.4 EM algorithm for Model 2

Derivation. For Model 2, $Y_n \mid X_n, \sigma^2 \sim \mathcal{N}(X_n, \sigma^2 \exp(X_n))$. The complete-data log-likelihood contribution from observations is

$$\ell_{\text{obs}}(\sigma^2) = -\frac{T+1}{2} \log \sigma^2 - \frac{1}{2} \sum_{n=0}^T X_n - \frac{1}{2\sigma^2} \sum_{n=0}^T \frac{(Y_n - X_n)^2}{\exp(X_n)}.$$

Setting $\partial \mathbb{E}[\ell_{\text{obs}} \mid y_{0:T}, \theta_k] / \partial \sigma^2 = 0$ gives the M-step:

$$\sigma_{k+1}^2 = \frac{1}{T+1} \mathbb{E}_{\theta_k} \left[\sum_{n=0}^T (Y_n - X_n)^2 \exp(-X_n) \mid y_{0:T} \right].$$

The required E-step quantity is a sum of marginal smoothing expectations, since each summand depends only on X_n . Thus the target can be written as $\sum_{n=0}^T \mathbb{E}_{\theta_k}[h_n(X_n) \mid y_{0:T}]$ with $h_n(x) = (Y_n - x)^2 \exp(-x)$; it does not intrinsically require a joint trajectory functional. Full trajectories are nevertheless a convenient Monte Carlo implementation route for both the genealogy-based SIR approximation and FFBSa.

Implementation. The E-step expectation is approximated in two ways:

- **SIR path:** use the particle genealogy to trace full trajectories and compute the weighted sum. This suffers from path degeneracy, the same mechanism seen in the additive-functional experiment.
- **FFBSa:** draw $M = N$ backward trajectories and average. This has higher computational cost ($O(N^2T)$ per iteration) but lower variance for the E-step.

Both are run for 100 iterations with $N = 100$ on dataset y^2 , from three initial values $\sigma_0^2 \in \{0.3, 0.5, 1.0\}$. Because the E-step is approximated by particles, this is a Monte Carlo EM implementation rather than deterministic EM; monotone, smooth convergence is therefore not expected from a single noisy trace.

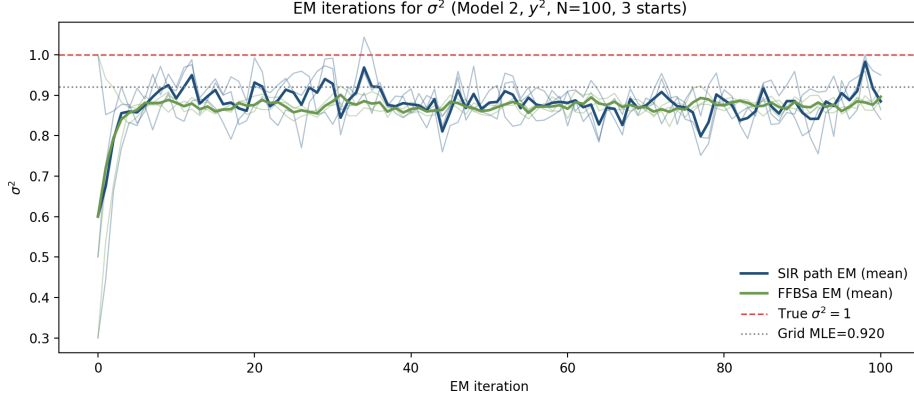


Figure 13: EM iteration traces for σ^2 using SIR path and FFBSa E-steps. Faint lines show individual initialisations and thick lines show the across-start mean. Red dashed: true $\sigma^2 = 1$; grey dotted: Model 2 grid MLE.

Monte Carlo EM behaviour. The two Monte Carlo EM versions differ only in how the E-step expectation is approximated. In the SIR-path version, this expectation is built from terminal genealogies and so inherits path-degeneracy noise; in the FFBSa version, it is approximated from backward-sampled smoothing trajectories, giving a lower-variance trajectory estimate. Across the three starts, the SIR-path final estimates are 0.950, 0.840, and 0.864 (mean 0.885, sd 0.058), while the FFBSa final estimates are 0.889, 0.919, and 0.880 (mean 0.896, sd 0.021). The FFBSa E-step is therefore more stable across starts and ends close to the Model 2 PF grid MLE of 0.920; remaining differences arise from Monte Carlo noise with $N = 100$, the larger $N = 500$ used in the grid likelihood, and the fact that $\log \hat{p}$ is biased even when $\hat{p}$ is unbiased.

3 Bayesian inference for a real data-set

Model

The stochastic volatility model for daily GBP/USD log returns $y_{1:T}$ is

$$X_n = \rho X_{n-1} + \sigma V_n, \quad V_n \sim \mathcal{N}(0, 1), \quad (5)$$

$$Y_n = \beta \exp(X_n/2) W_n, \quad W_n \sim \mathcal{N}(0, 1), \quad (6)$$

$$X_0 \sim \mathcal{N}(0, 0.01). \quad (7)$$

Static parameters: $\theta = (\rho, \sigma^2, \beta^2)$. We use $T = 1000$ data points for fitting and the remaining ≈ 250 for prediction.²

3.1 Algorithm

I use **Particle Marginal Metropolis-Hastings (PMMH)**. The algorithm targets the posterior $p(\rho, \sigma^2, \beta^2 \mid y_{1:T})$ by plugging a particle filter estimate of the marginal likelihood $\hat{p}_\theta^N(y_{1:T})$ into a Metropolis-Hastings acceptance ratio. With multinomial resampling at every step, the bootstrap PF likelihood estimator is nonnegative and unbiased, which guarantees that the PMMH chain has the correct stationary distribution.

Priors. $\rho \sim \text{Uniform}(-1, 1)$, $\sigma^2 \sim \text{IG}(2, 0.02)$, $\beta^2 \sim \text{IG}(2, 10^{-4})$. The uniform prior on ρ encodes stationarity ($|\rho| < 1$) without further shape assumptions. I use *separate* inverse-gamma priors for σ^2 and β^2 because these two variance parameters live on very different scales in daily-return data:

²The assignment text later writes τ^2 for the latent innovation variance; since the model equation here uses σ , I report this variance as σ^2 throughout.

σ^2 controls latent log-volatility innovations and can reasonably be of order 10^{-2} , whereas β^2 is tied directly to the observed return scale and is expected to be much smaller. In short exploratory PMMH runs, using the same diffuse prior for both variances led to unrealistically wide predictive bands, whereas these scale-aware priors gave materially better predictive calibration while leaving the PMMH framework unchanged.

Transformed parameterisation. The MCMC chain runs on $(\eta_\rho, \log \sigma^2, \log \beta^2)$ where $\rho = \tanh(\eta_\rho)$. The Jacobian $|\partial(\rho, \sigma^2, \beta^2)/\partial(\eta, \log s, \log b)| = (1 - \rho^2) \sigma^2 \beta^2$ is included in the acceptance ratio.

A bootstrap particle filter with $N = 1000$ particles and multinomial resampling at every step estimates $\hat{p}_\theta^N(y_{1:T})$. Multinomial resampling ensures the likelihood estimator is unbiased, which is required for PMMH validity. The previous likelihood estimate is stored and reused on rejection (standard PMMH).

3.2 Implementation and diagnostics

A pilot PMMH run of 3,000 iterations with a diagonal Gaussian random-walk proposal (step sizes 0.01, 0.08, 0.08 on the transformed scale) is used to estimate the posterior scale and correlation. The main PMMH run then uses a fixed multivariate Gaussian random-walk proposal whose covariance is a scaled version of the empirical covariance from the second half of the pilot chain. To improve Monte Carlo precision while keeping the same target distribution, I combine five independent post-burn-in PMMH chains: four chains with 30,000 main iterations and burn-in 7,000, and one longer chain with 60,000 main iterations and burn-in 10,000. All use $N = 1000$ particles and the same priors; only the random seed and final proposal scale differ.

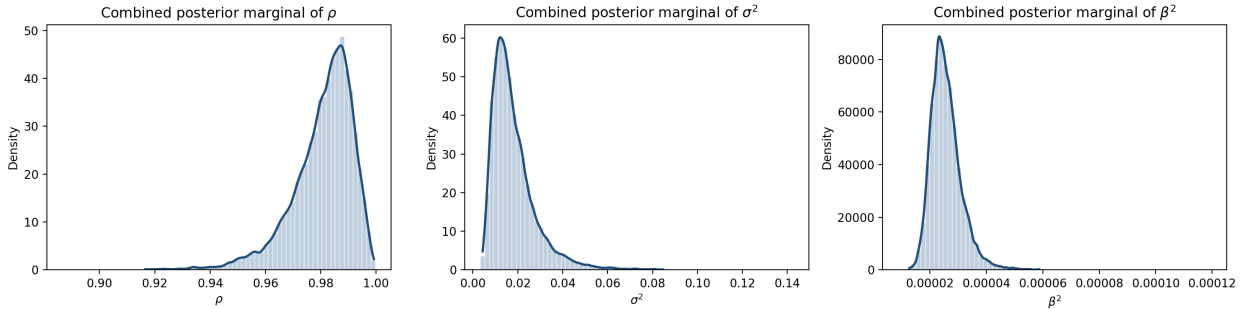


Figure 14: Posterior marginal densities for ρ , σ^2 , and β^2 .

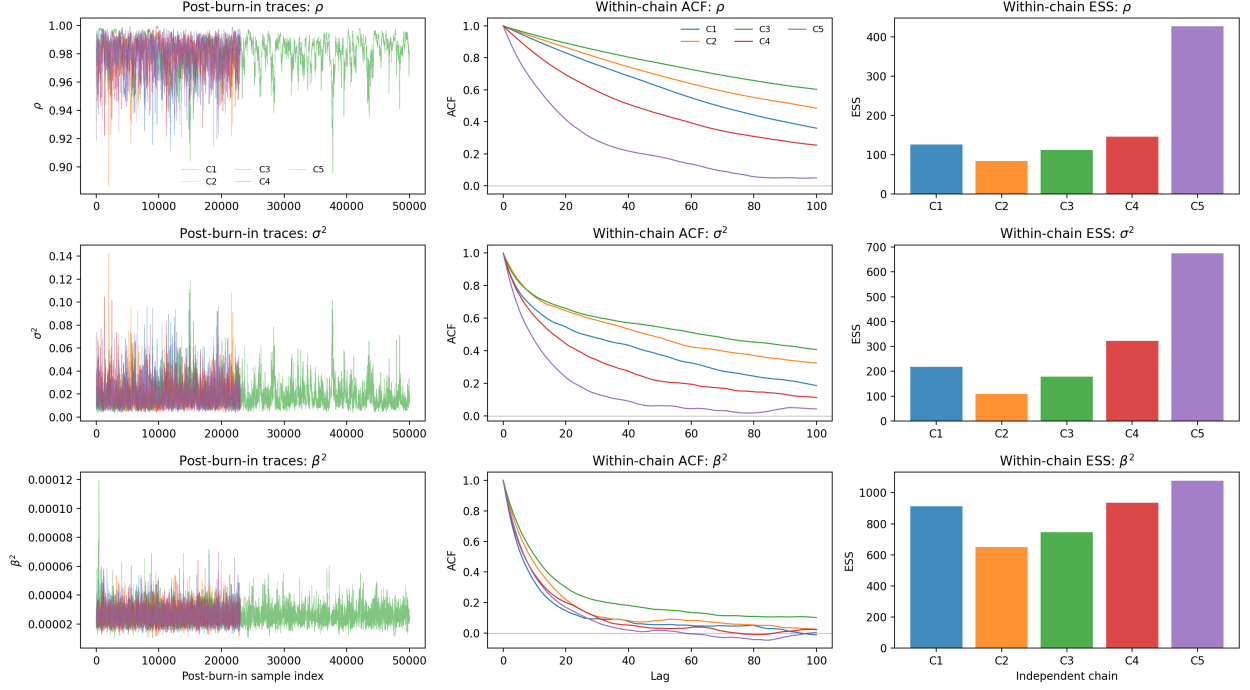


Figure 15: Post-burn-in trace plots, within-chain ACFs, and within-chain ESS for each parameter across five independent PMMH chains, labelled C1 to C5.

Table 3: Combined PMMH posterior summary. ESS values are sums of within-chain ESS over five independent post-burn-in chains.

	ρ	σ^2	β^2
Posterior mean	0.981	0.0181	2.57×10^{-5}
Posterior std	0.0112	0.0101	5.62×10^{-6}
95% CI	[0.953, 0.996]	[0.0064, 0.0445]	$[1.74 \times 10^{-5}, 3.85 \times 10^{-5}]$
ESS	895.7	1501.2	4321.6
Acceptance rates	21.6%, 30.2%, 26.0%, 23.4%, 26.8%		

Diagnostics. The acceptance rates, between 21.6% and 30.2%, are within the usual effective range for pseudo-marginal random-walk samplers. The posterior scale is plausible for daily returns: β^2 is of order 10^{-5} , in line with the observed data scale, and the predictive intervals are no longer excessively inflated. Mixing is slowest for ρ and σ^2 because of the posterior ridge between persistence and innovation variance; the ACF panels in Figure 15 show the slow decay directly. The pilot-calibrated covariance proposal accounts for this correlation, but it cannot remove the intrinsic slow movement along the ridge. Combining independent chains gives a larger effective sample size: the total within-chain ESS is 895.7 for ρ , 1501.2 for σ^2 , and 4321.6 for β^2 . Figure 16 shows that the five independent chains have overlapping marginal posterior shapes and close posterior means, so the combined posterior summaries in Table 3 are not driven by a single run. The cost is $O(NT) = 10^6$ per iteration.

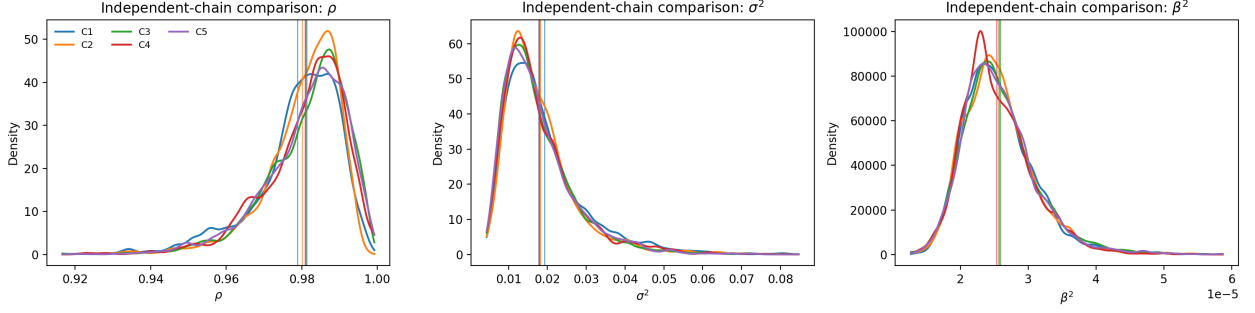


Figure 16: Posterior marginal comparison from five independent PMMH chains, labelled C1 to C5. Vertical lines mark chain-specific posterior means.

3.3 Predictive posterior

The posterior predictive distribution is

$$p(y_{T+1:T'} | y_{1:T}) = \int p(y_{T+1:T'} | x_T, \theta) p(x_T | y_{1:T}, \theta) p(\theta | y_{1:T}) dx_T d\theta.$$

I approximate this integral as follows:

1. Draw M thinned posterior samples $\vartheta^{(m)} = (\rho^{(m)}, (\sigma^2)^{(m)}, (\beta^2)^{(m)})$ from the combined post-burn-in chains.
2. Define $\sigma^{(m)} = \sqrt{(\sigma^2)^{(m)}}$ and $\beta^{(m)} = \sqrt{(\beta^2)^{(m)}}$. For each $\vartheta^{(m)}$, run the particle filter on $y_{1:T}$ to obtain an approximate sample $X_T^{(m)} \sim p(x_T | y_{1:T}, \vartheta^{(m)})$.
3. Propagate forward: $X_n^{(m)} = \rho^{(m)} X_{n-1}^{(m)} + \sigma^{(m)} V_n$ for $n > T$.
4. Generate predictive observations: $Y_n^{(m)} = \beta^{(m)} \exp(X_n^{(m)}/2) W_n$.
5. Compute pointwise quantiles and mean across $m = 1, \dots, M$.

This marginalises over both parameter uncertainty (through posterior samples) and state uncertainty (through the PF and forward simulation), giving a fully Bayesian predictive distribution. For simplicity, I rerun the particle filter for the retained posterior draws instead of storing terminal particle clouds during PMMH.³

3.4 Predictive intervals

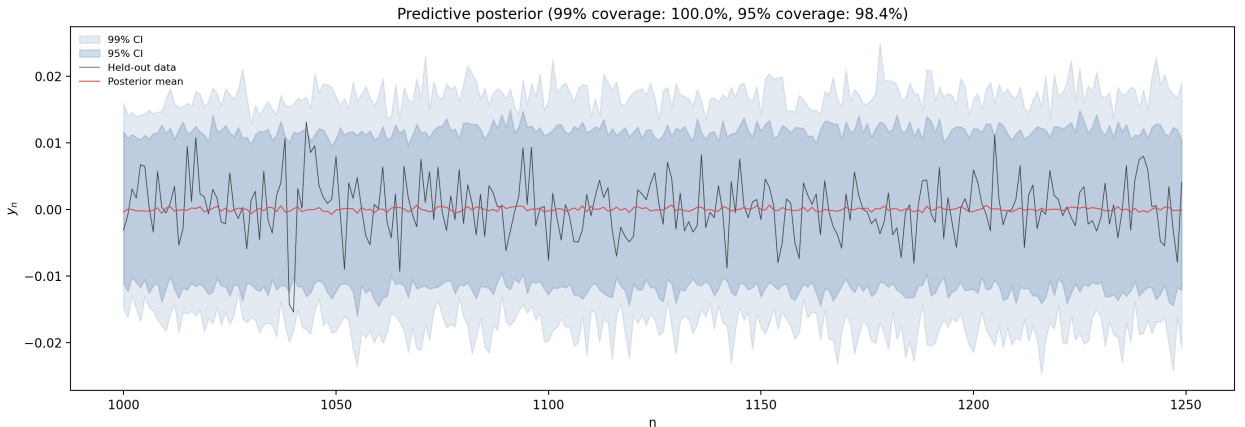


Figure 17: Predictive posterior intervals vs. held-out data. The 99% credible intervals use the 0.005 and 0.995 quantiles. Coverage statistics are shown in the title.

³Storing terminal particle clouds from accepted PMMH states would reduce computation but would require extra memory and bookkeeping.

I use 99% intervals (0.005, 0.995 quantiles) and additionally report 95% coverage.⁴

3.5 Model assessment

Strengths. The stochastic volatility model captures several stylised facts of financial returns:

- **Volatility clustering:** persistent latent X_n (high ρ) generates periods of high and low volatility.
- **Heavy tails:** marginalising over X_n produces heavier-than-Gaussian predictive distributions for Y_n .
- **Bayesian uncertainty:** parameter uncertainty is propagated into predictions, although in this implementation the resulting intervals are conservative (see below).

Limitations.

- **Zero conditional mean:** the model assumes $\mathbb{E}[Y_n | X_n] = 0$, ignoring potential drift in returns.
- **No leverage effect:** there is no correlation between V_n and W_n , so negative returns do not increase future volatility (the “leverage effect” observed in equity markets).
- **Gaussian innovations:** both state and observation shocks are Gaussian; real returns may have fatter tails even conditional on the volatility state.
- **AR(1) dynamics:** more complex volatility dynamics (e.g. two-factor, long memory) might be needed.
- **No jumps or structural breaks:** sudden regime changes or event-driven jumps are not modelled.

The empirical coverage of the predictive intervals is reported in the figure title. Here the 95% coverage is 98.4% and the 99% coverage is 100%, so the predictive distribution is still somewhat conservative, especially in the extreme tails. This is partly because the Bayesian prediction integrates over parameter uncertainty, including the slowly-mixing ρ marginal, which inflates predictive variance. Additionally, the high posterior persistence ($\hat{\rho} \approx 0.981$) means predictive volatility reverts only gradually, keeping the intervals wide. Finally, with only ~ 250 held-out points, 100% empirical coverage at the 99% level is not surprising: the expected number of exceedances is only ~ 2.5 . The prediction intervals are mildly conservative rather than excessively wide. This is acceptable for risk-style prediction, but it also indicates that the model still slightly overestimates uncertainty, particularly for tail events.

The stochastic volatility model is reasonably appropriate for this data-set: it captures heteroscedasticity and produces credible predictive bands with sensible scale, but its simplicity (zero mean, Gaussian shocks, no leverage or jumps, AR(1) log-volatility) still limits fine-grained fit and leaves some conservatism in the predictive distribution.

⁴The question also mentions 0.025 and 0.975 quantiles, which correspond to 95%; reporting both coverages avoids relying on only one reading.